# Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau

**Project Report**

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## Team Members


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## Tools Used

- Tableau
- Microsoft Excel
- GitHub

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## 1. Introduction

Electricity consumption is an important indicator of economic growth and development. With increasing industrialization, urbanization, and population growth, electricity demand has been rising rapidly. Understanding electricity consumption patterns helps policymakers and organizations improve energy management and plan better energy distribution.

This project focuses on analyzing electricity consumption data and presenting insights using interactive dashboards. By using modern data visualization techniques, complex datasets can be transformed into meaningful visual insights that help identify trends and patterns.

The project uses Tableau to create interactive dashboards that allow users to analyze electricity consumption across different states and regions in India.

# 2. Ideation Phase

## 2.1 Problem Statement

Electricity consumption data is often large and complex, making it difficult to interpret trends and patterns. Without proper visualization tools, understanding how electricity usage changes across time and regions becomes challenging.

The objective of this project is to analyze electricity consumption data and transform it into visual dashboards that allow users to explore trends, compare states, and observe regional patterns.

## 2.2 Empathy Map Canvas

Stakeholders such as energy planners, government authorities, and researchers require clear insights into electricity usage patterns. They need tools that allow them to easily analyze consumption trends and understand how external factors affect electricity demand.

This project helps stakeholders by presenting electricity consumption data through interactive visual dashboards.

## 2.3 Brainstorming

During the brainstorming phase, several approaches for analyzing electricity data were considered. Data visualization was chosen as the best method to present insights in a clear and interactive manner.

Among different visualization tools, **Tableau** was selected because it allows the creation of dynamic dashboards, maps, and charts that simplify complex datasets.

# 3. Requirement Analysis

## 3.1 Dataset

The dataset used for this project is **Consumption.csv**, which contains electricity consumption information across various states and time periods.

The dataset includes information related to electricity usage patterns that can be analyzed to identify trends and regional differences.

## 3.2 Technology Stack

| Tool | Purpose |
|---|---|
| Tableau | Data visualization and dashboard creation |
| Microsoft Excel | Data preprocessing |
| GitHub | Version control and project repository |

The project repository is hosted on **GitHub**.

# 4. Project Design

## 4.1 Problem–Solution Fit

The project addresses the challenge of understanding electricity consumption patterns by transforming raw data into visual dashboards. Visualization makes it easier to interpret complex datasets and identify trends in electricity usage.

## 4.2 Proposed Solution

The proposed solution is to analyze electricity consumption data and create dashboards that display consumption trends, regional distribution, and state-wise comparisons.

These dashboards allow users to explore electricity data interactively and gain insights into energy usage patterns.

### 4.3 Solution Architecture

The workflow of the project includes the following steps:

Dataset Collection → Data Cleaning → Data Visualization → Dashboard Creation → Insight Generation

## 5. Project Planning and Scheduling

| Phase | Task |
|---|---|
| Week 1 | Data collection |
| Week 2 | Data preprocessing |
| Week 3 | Dashboard creation |
| Week 4 | Documentation |

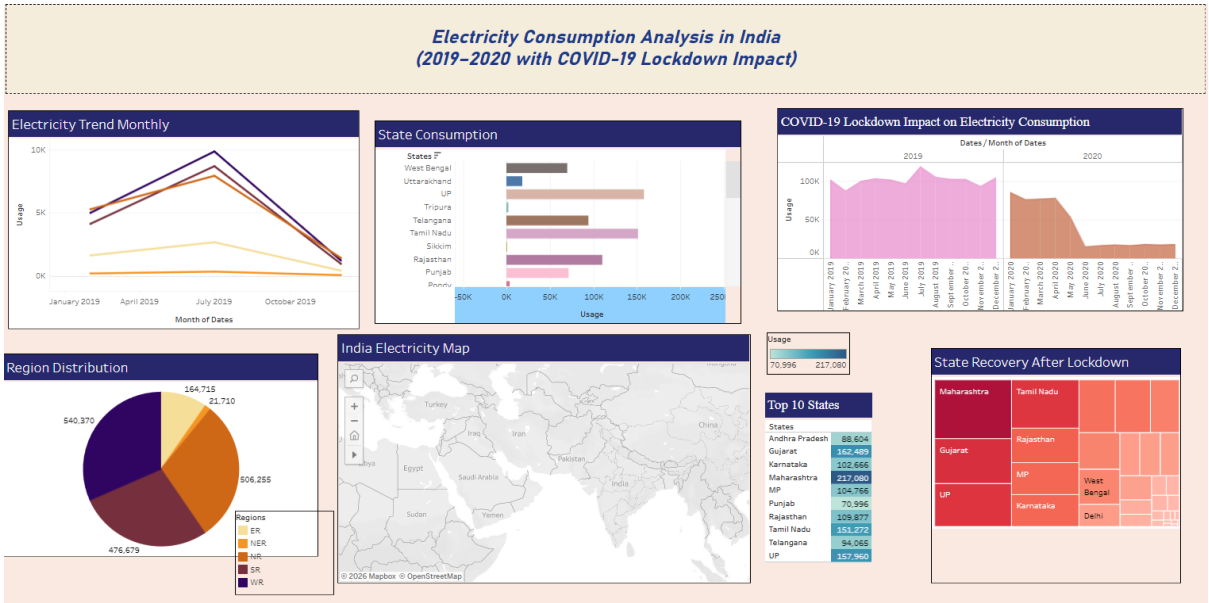
## 6. Functional and Performance Testing

The dashboards were tested to ensure that all visualizations display accurate data. Filters and visual elements were checked to ensure they respond correctly to user interactions.

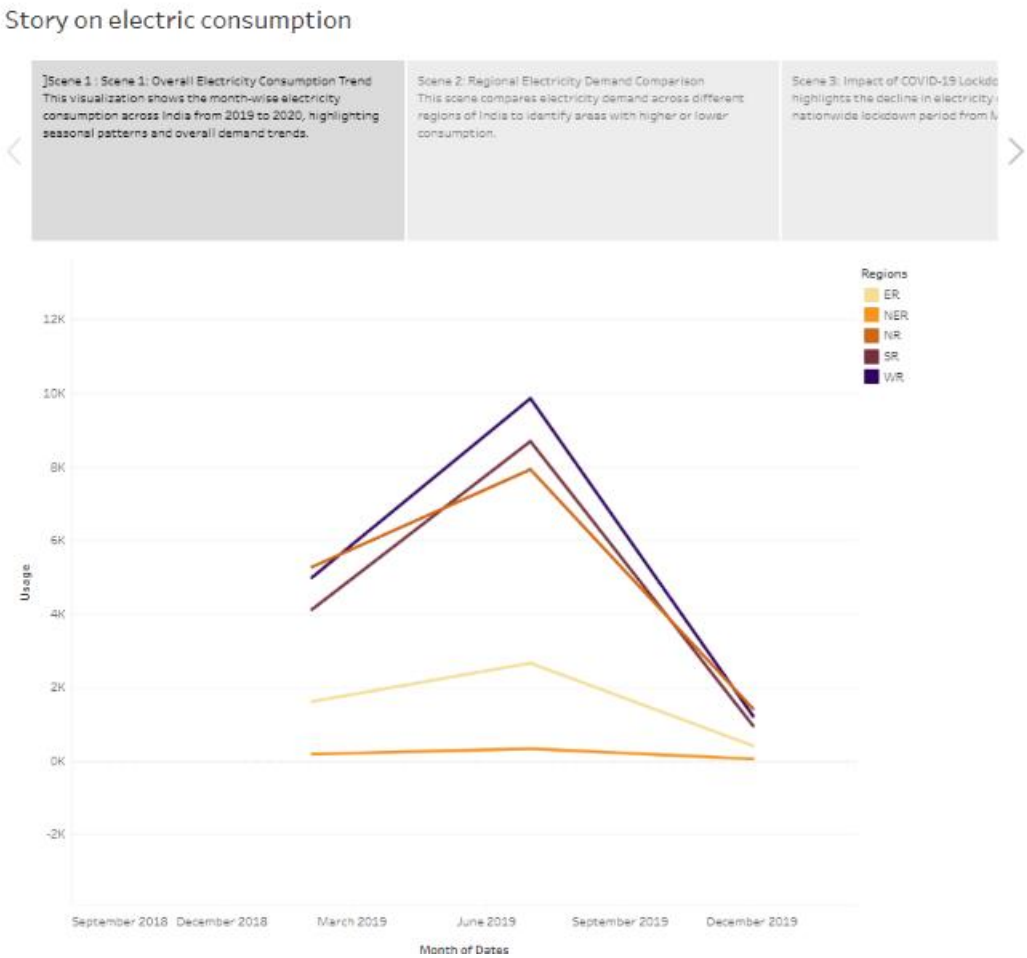
The dashboard performance was evaluated to ensure smooth interaction and proper display of information.

## 7. Results

### Dashboard Visualization




# STORY SCREENSHOTS

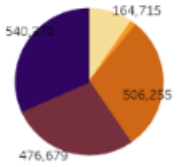


## Story on electric consumption

]Scene 1 : Scene 1: Overall Electricity Consumption Trend
This visualization shows the month-wise electricity consumption across India from 2019 to 2020, highlighting seasonal patterns and overall demand trends.

Scene 2: Regional Electricity Demand Comparison
This scene compares electricity demand across different regions of India to identify areas with higher or lower consumption.

Scene 3: Impact of COVID-19 Lockdo
highlights the decline in electricity
nationwide lockdown period from M




Regions
- ER
- NER
- NR
- SR
- WR

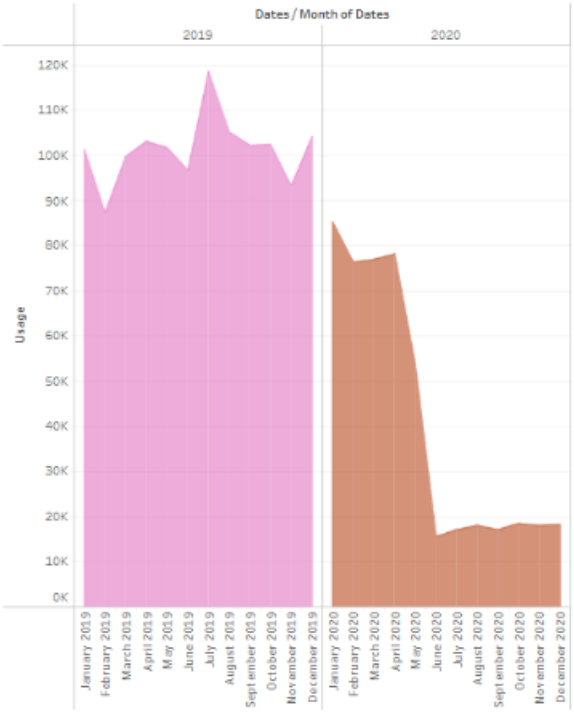
## Story on electric consumption

ity Consumption Trend
h-wise electricity
19 to 2020, highlighting
and trends.

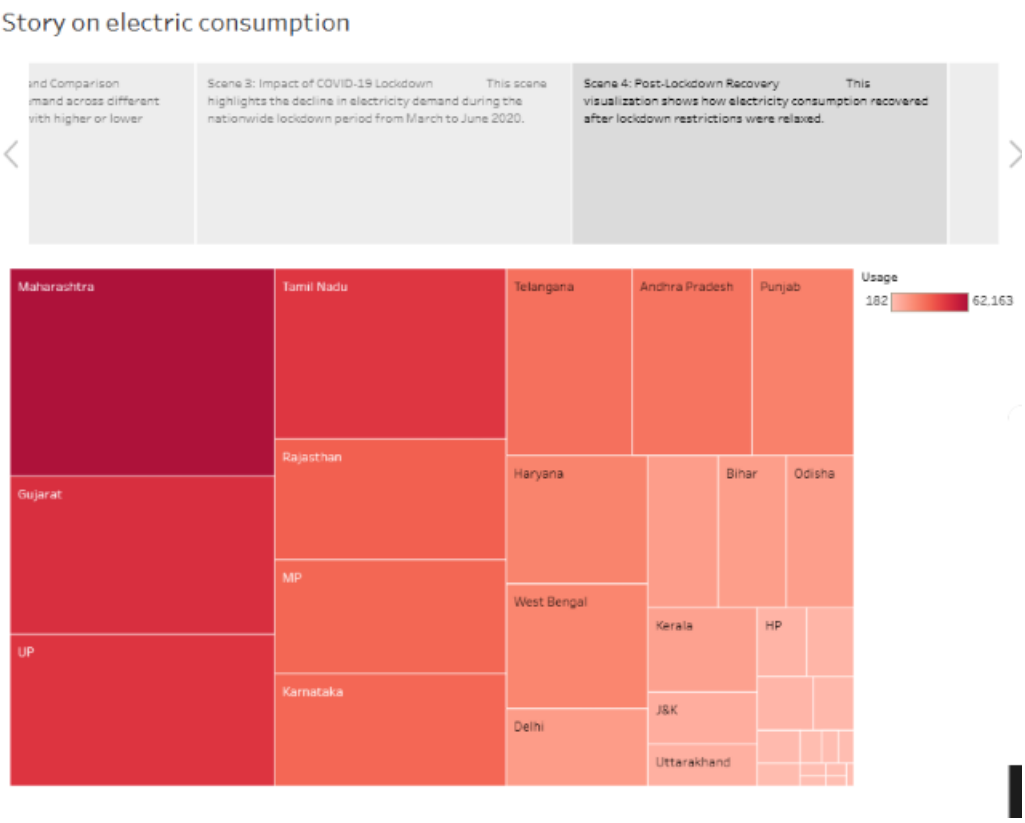
Scene 2: Regional Electricity Demand Comparison
This scene compares electricity demand across different regions of India to identify areas with higher or lower consumption.

Scene 3: Impact of COVID-19 Lockdown This scene highlights the decline in electricity demand during the nationwide lockdown period from March to June 2020.

Scene-
visuali
after l




Year of Dates
- 2019
- 2020




The dashboard presents electricity consumption analysis across different states and regions of India.

It includes several visualizations such as:

- Electricity Trend Monthly
- State Consumption Analysis
- COVID-19 Lockdown Impact
- Region Distribution
- India Electricity Map
- Top 10 Electricity Consuming States
- State Recovery After Lockdown

These visualizations help identify electricity consumption trends and analyze the impact of external events such as lockdowns on electricity usage.

## 8. Advantages and Disadvantages

### Advantages

- Simplifies complex electricity datasets
- Interactive dashboards allow better data exploration
- Helps identify electricity consumption trends
- Provides clear visual insights

### Disadvantages

- Results depend on dataset quality
- Visualization provides descriptive insights rather than predictions

## 9. Conclusion

This project successfully analyzes electricity consumption patterns using interactive dashboards created with Tableau.

The dashboards help visualize electricity usage trends, compare state-wise consumption, and analyze regional distribution patterns.

The insights generated through this project can help policymakers and researchers understand electricity demand and plan energy resources more effectively.

## 10. Future Scope

Future improvements to this project may include:

- Real-time electricity consumption monitoring
- Integration with machine learning models for prediction
- Advanced dashboards with automated data updates
- Smart energy planning tools

## 11. Appendix

### Tableau Public Dashboard

https://public.tableau.com/views/India_Electricity_Consumption_Dashboard/Dashboard1 Published on **Tableau Public**

## GitHub Repository

https://github.com/moana-pri/Electricity-Consumption-Analysis

Hosted on **GitHub**